

Goal

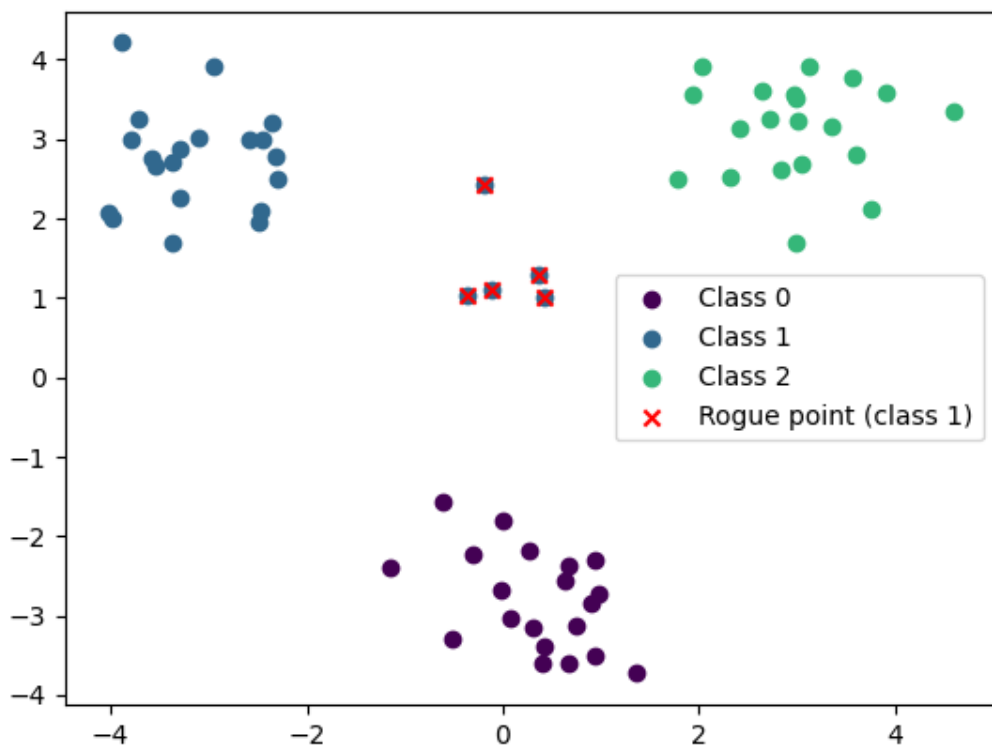
In this experiment we want to investigate the behavior of the SSD method and how the choice of hyperparameters affect the decision boundary. We also try to address the shortcomings of SSD in two ways:

- 1. By applying dampening at only the last layer of the model. This is referred to as "SSD v2"
- 2. By searching for hyperparameters whose dampening yields a loss as close as possible to the original model's loss on a "generalization dataset" constructed in the same way as the repair step for Scrub+R. This is referred to as "SSD v3".
- 3. A combination of the previous two approaches. This is referred to as "SSD v4".

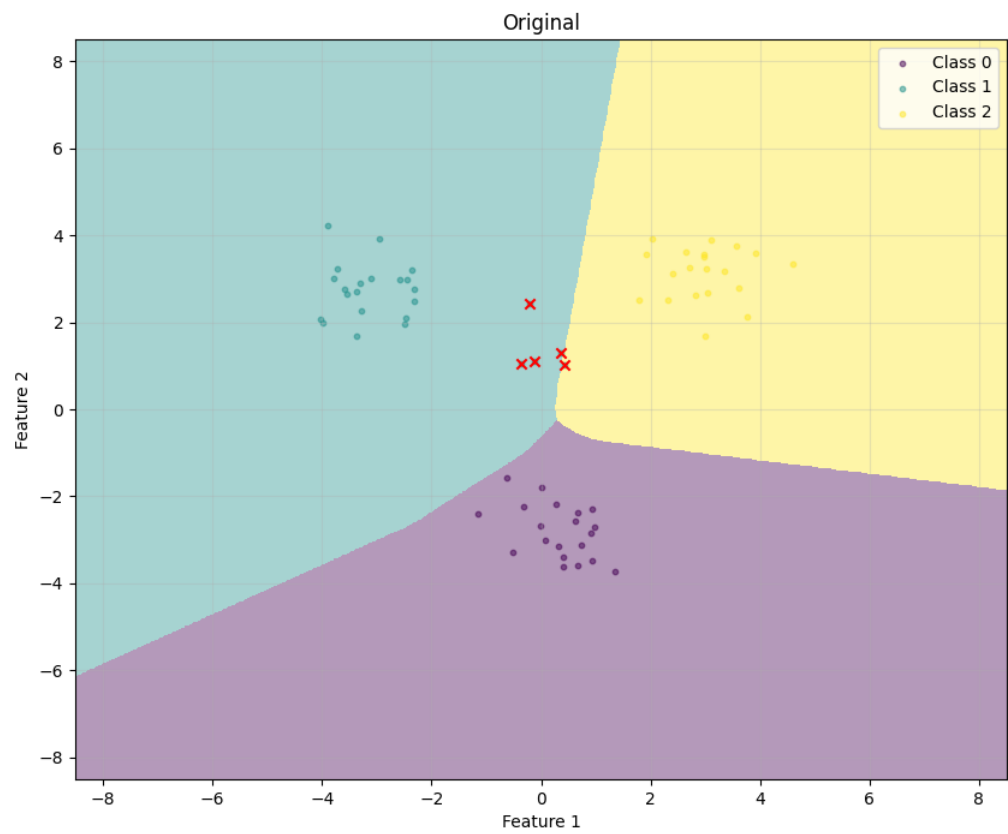
Experiment runs

Now we run the experiment on each of the dataset configuration, moving the rogue point.

1. Rogue point with same distance to all centroids (close to decision boundary)



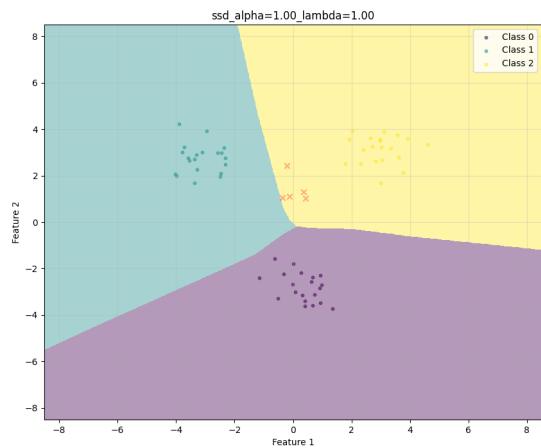
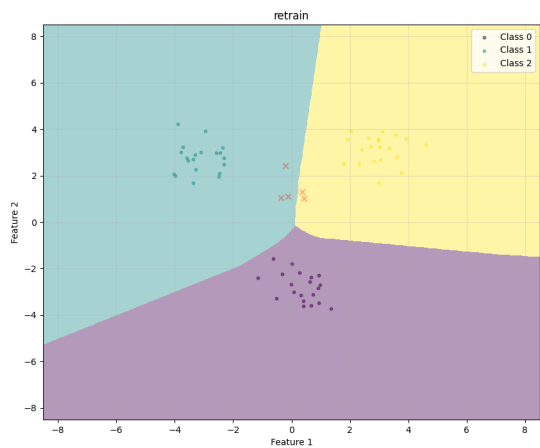
Original model for the dataset

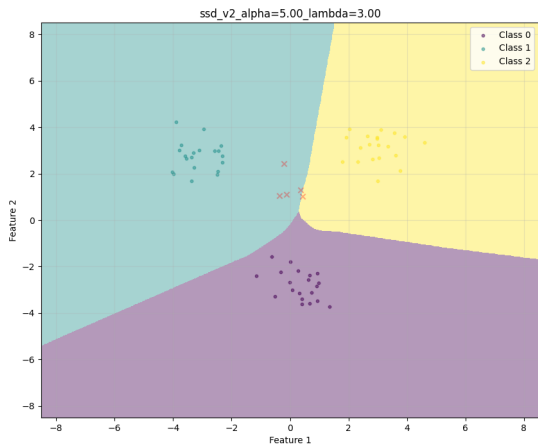
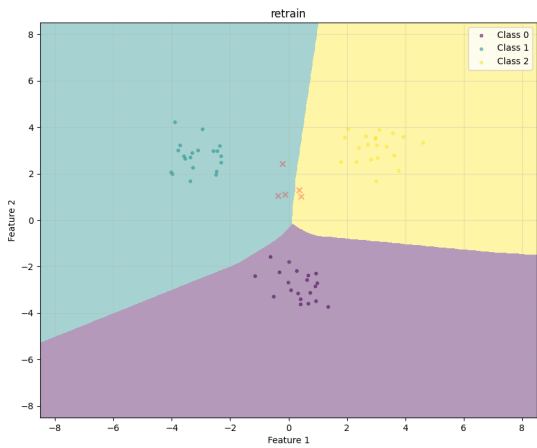
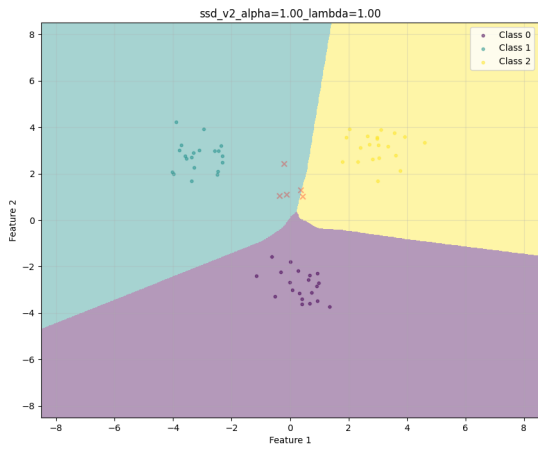
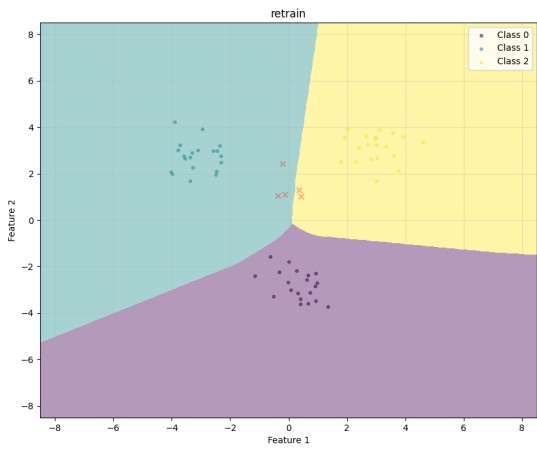
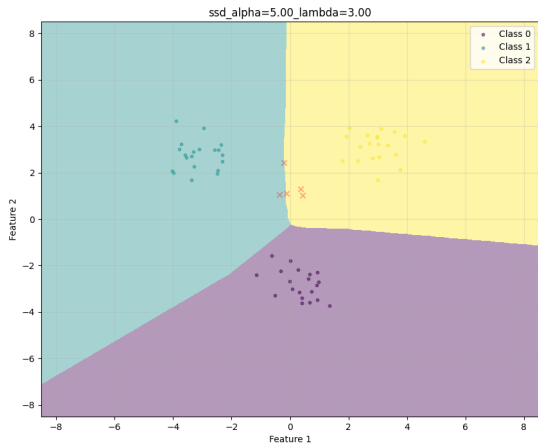
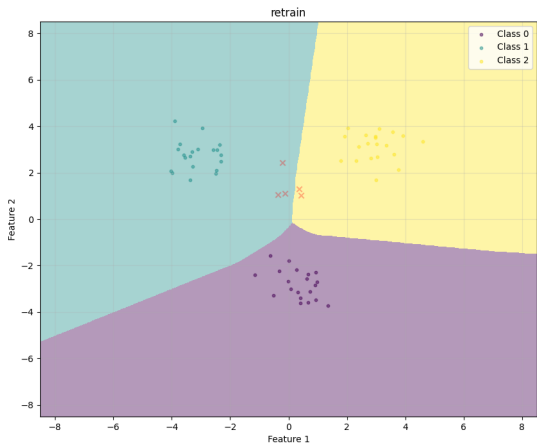


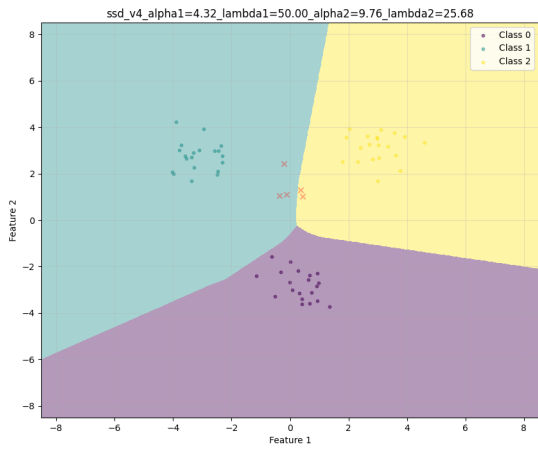
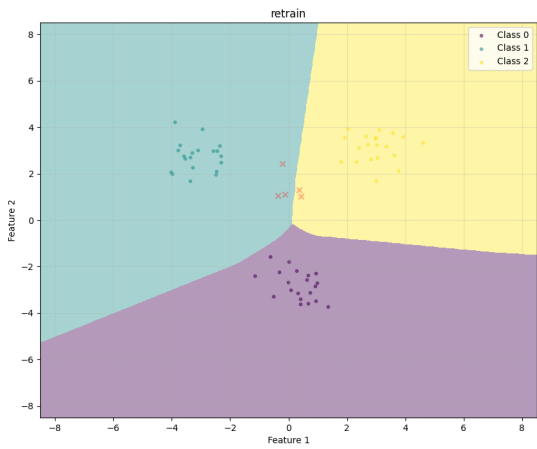
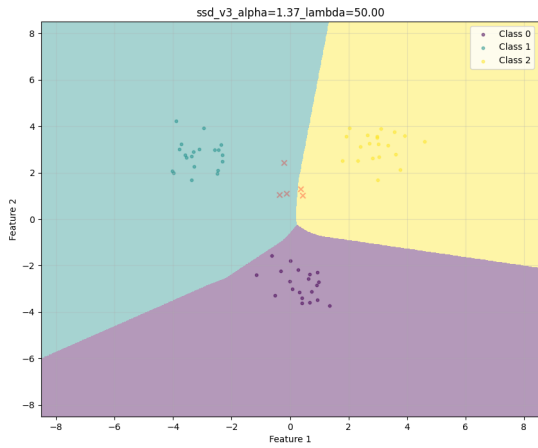
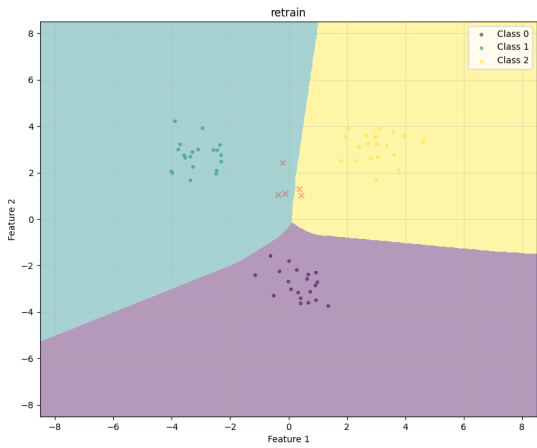
Findings

Unlearned

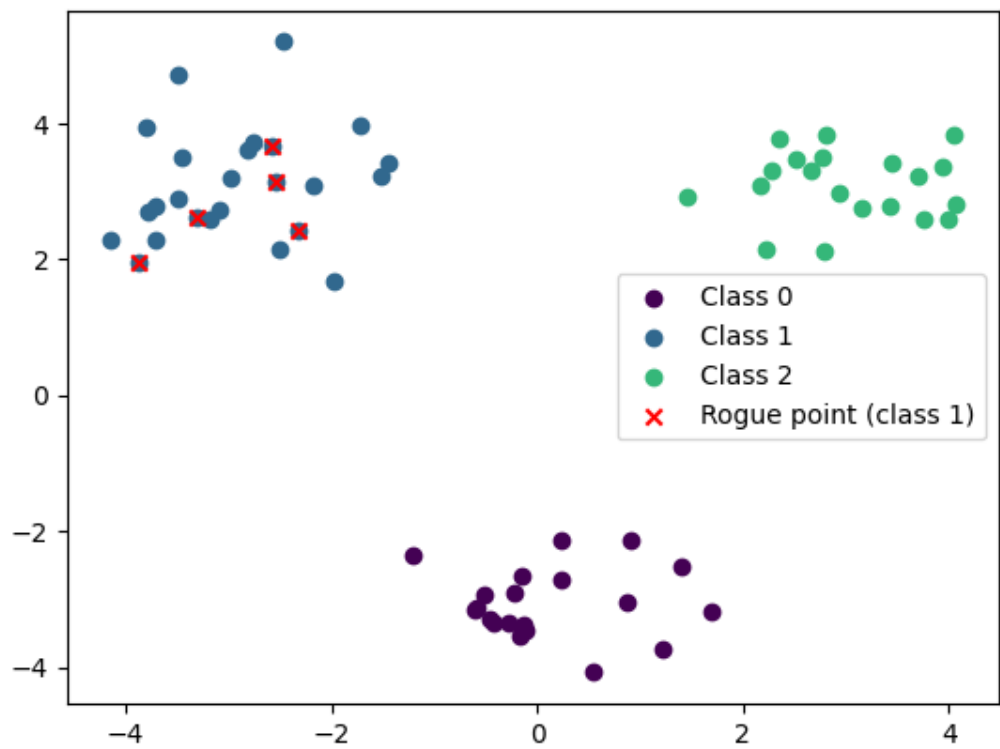
SSD



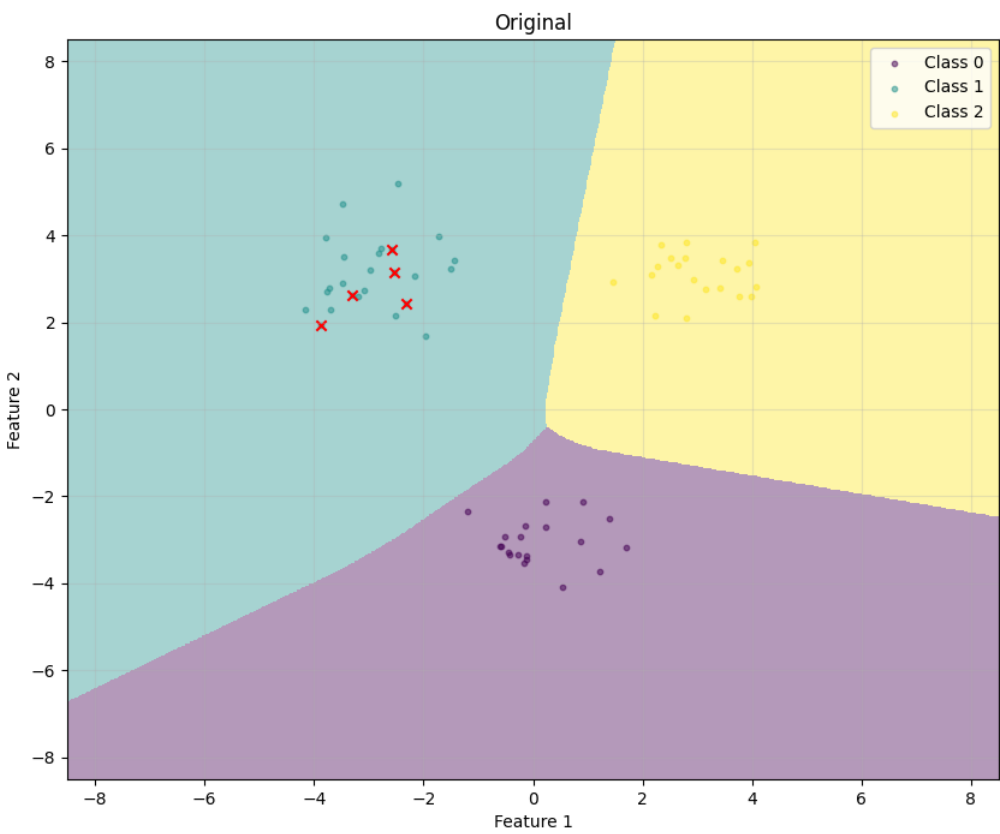




2. Rogue point with same centroid as its class



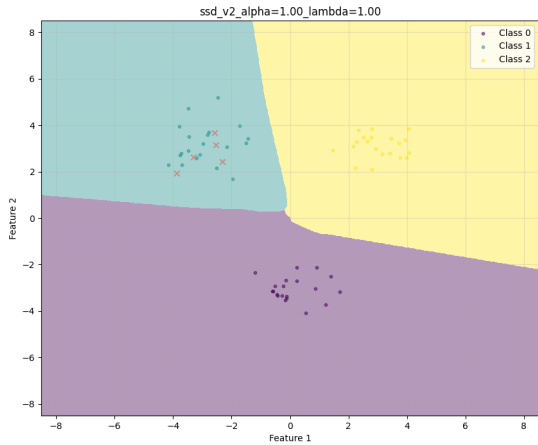
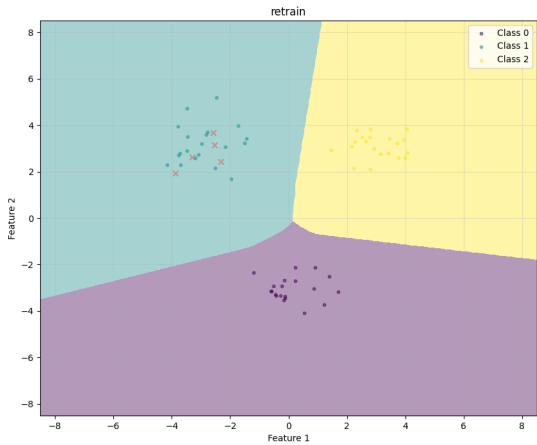
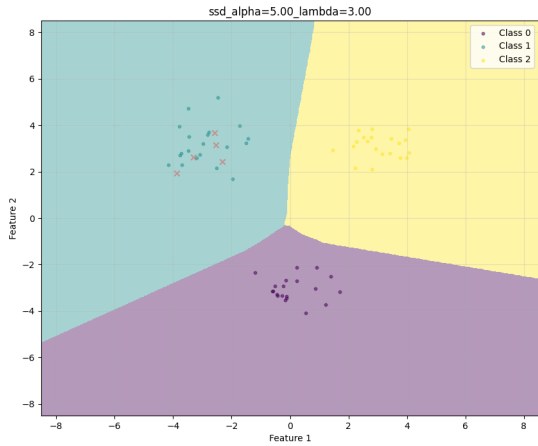
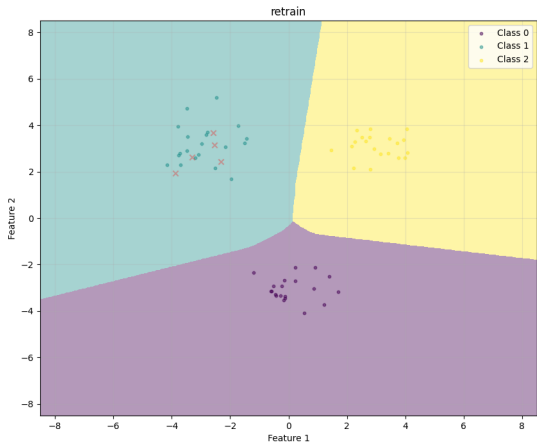
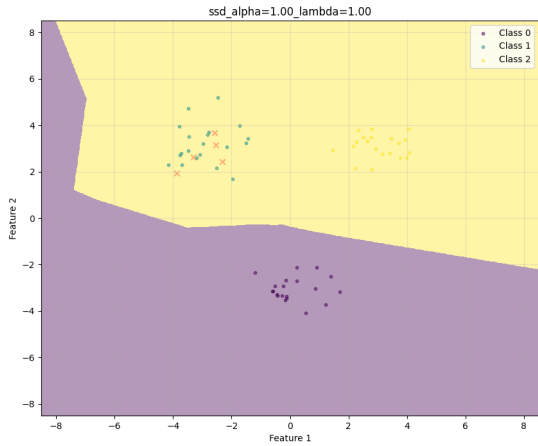
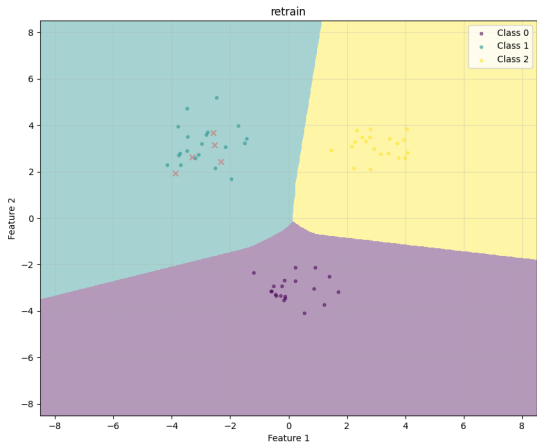
Original model for the dataset

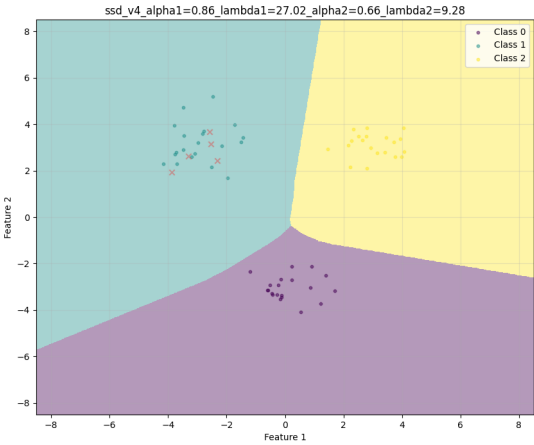
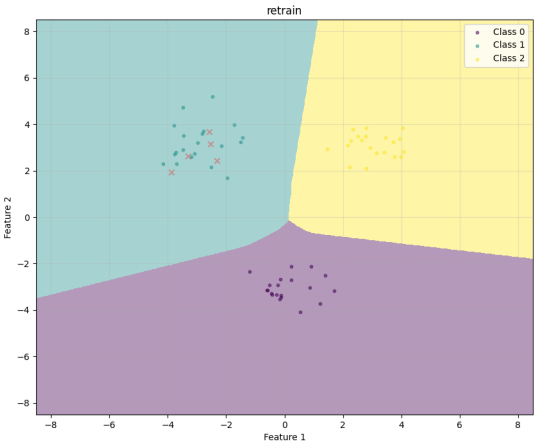
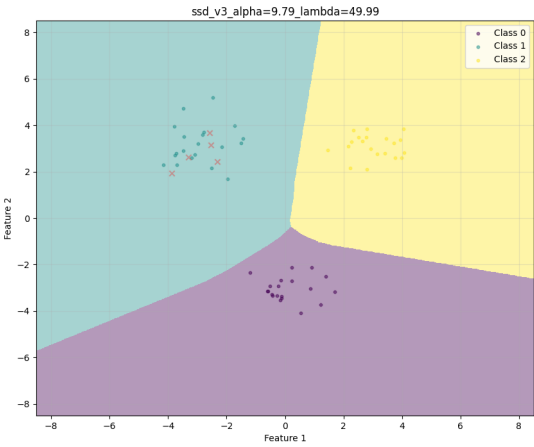
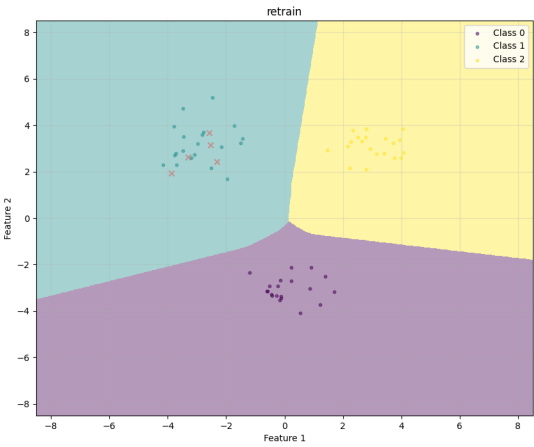
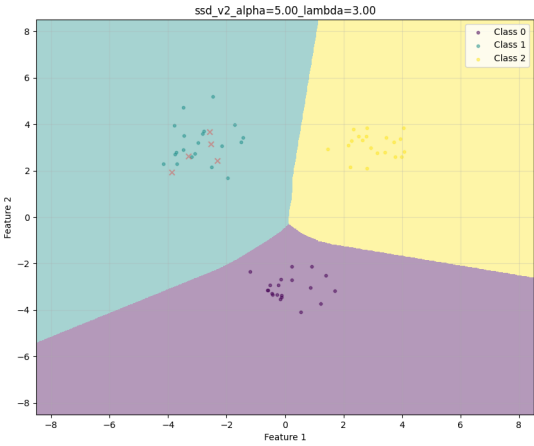
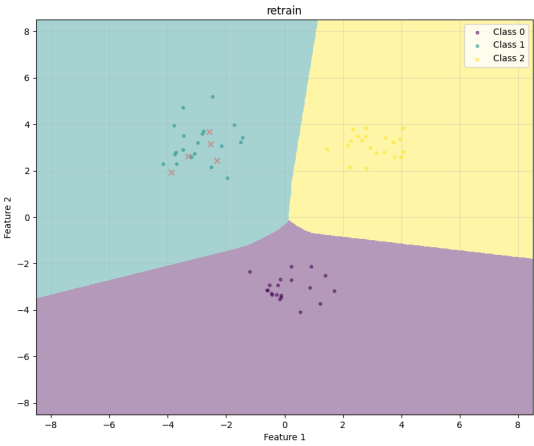


Findings

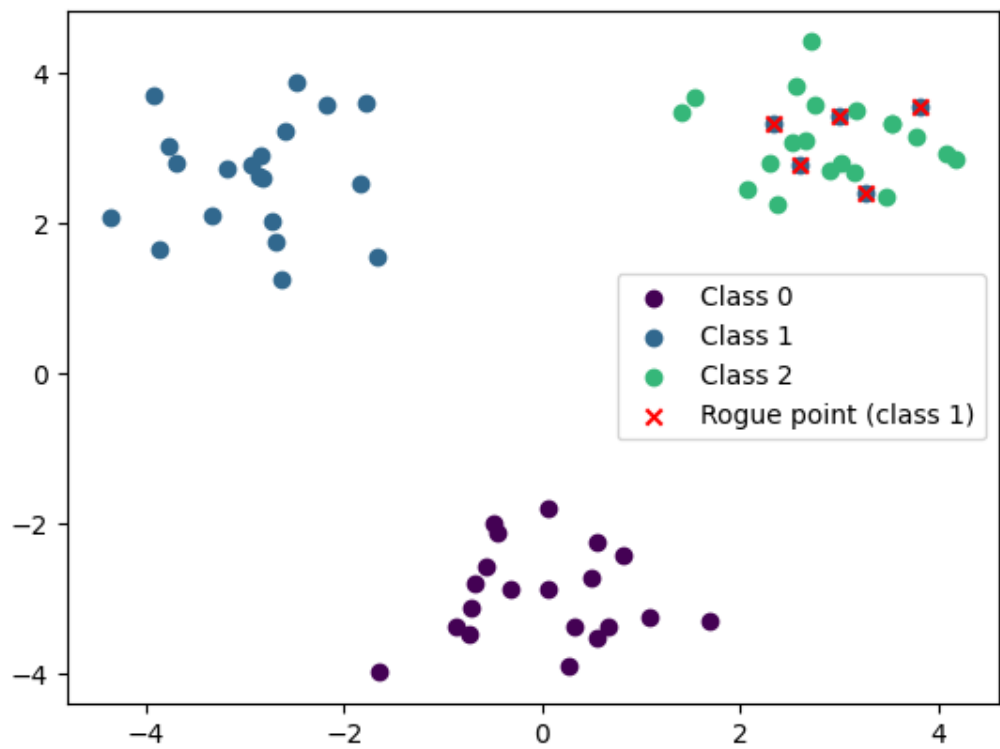
Unlearned

SSD

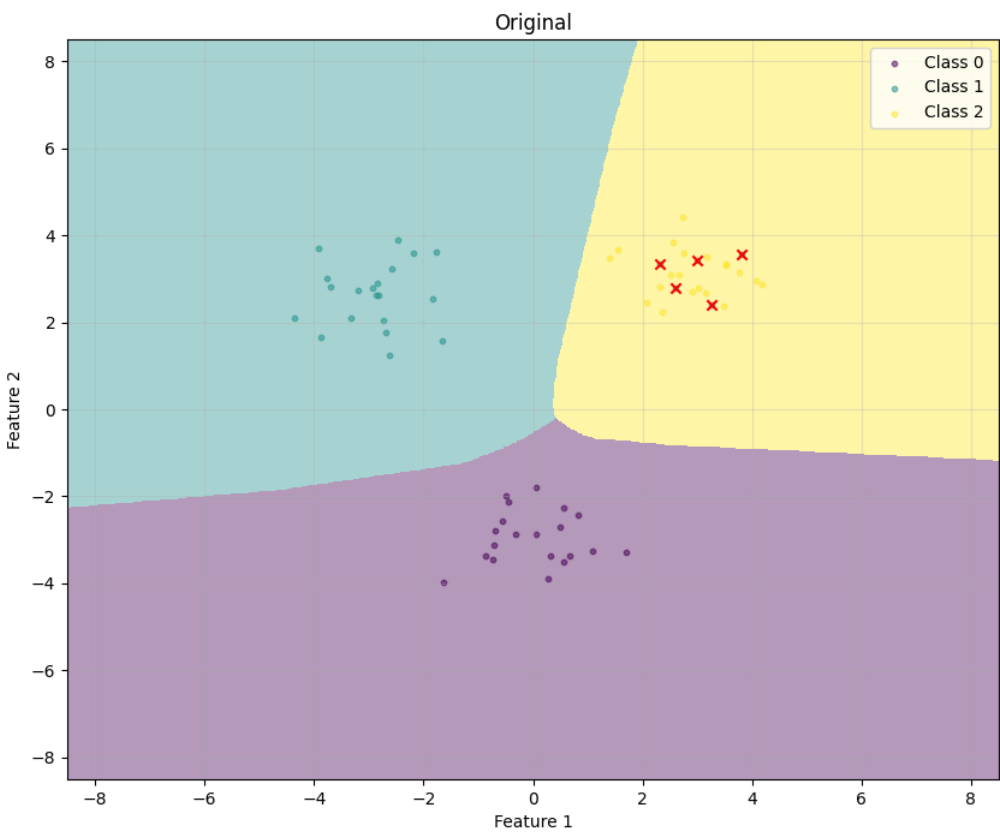




3. Rogue point with same centroid as the class with a different label



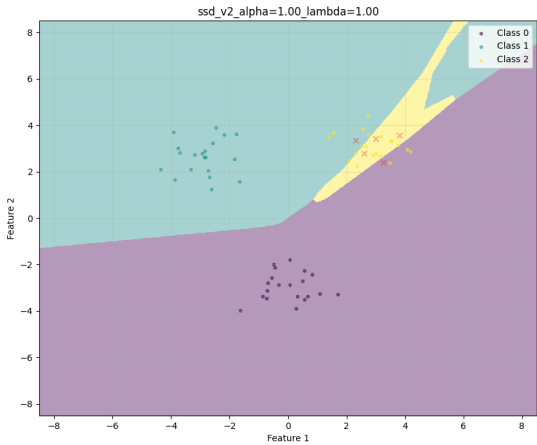
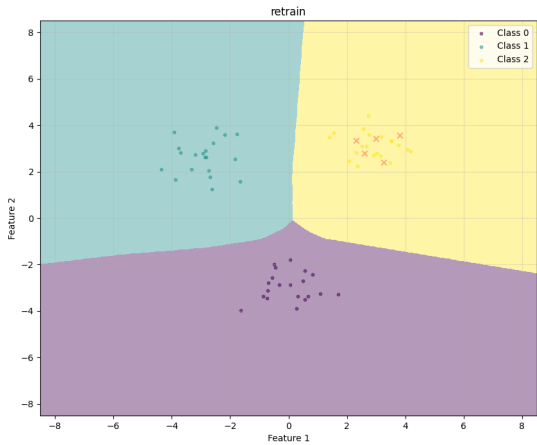
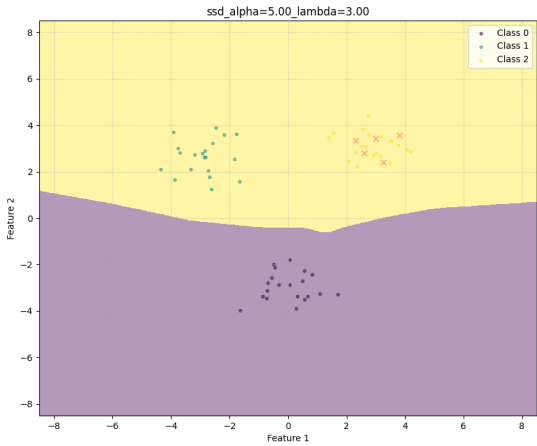
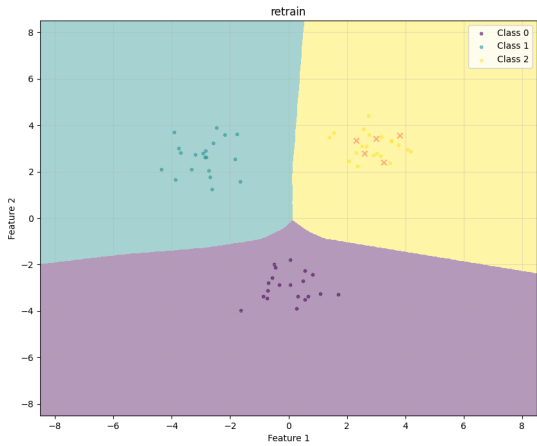
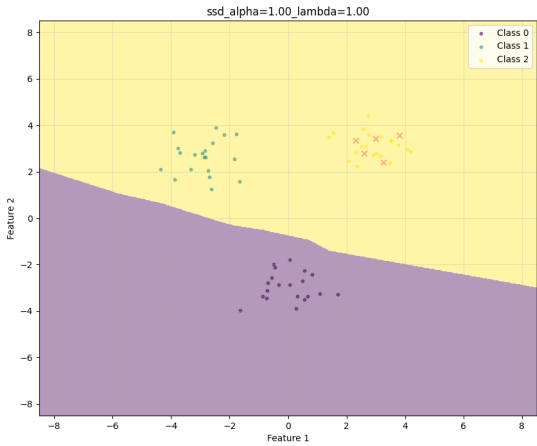
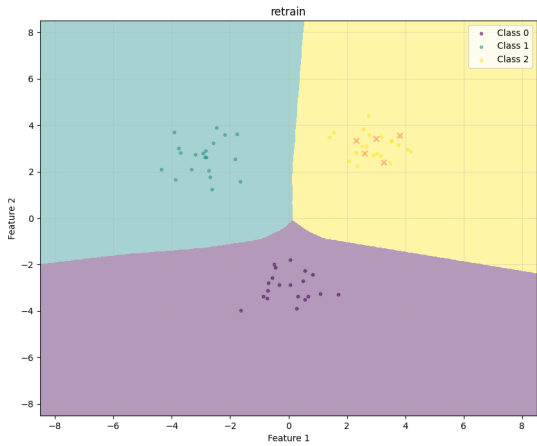
Original model for the dataset

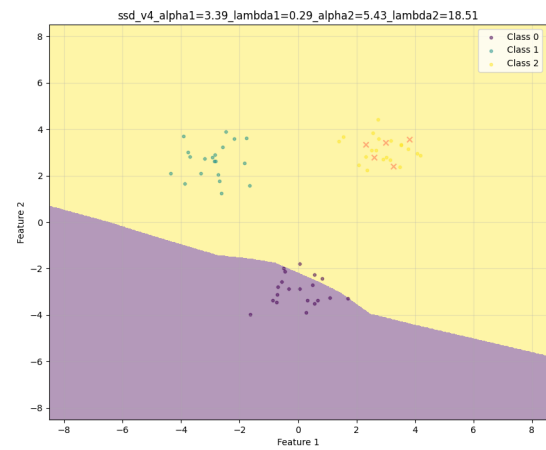
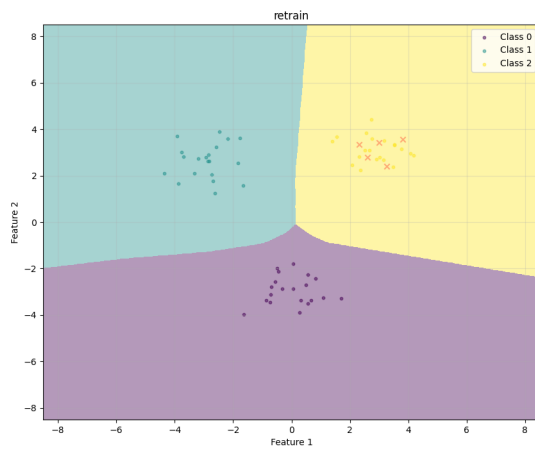
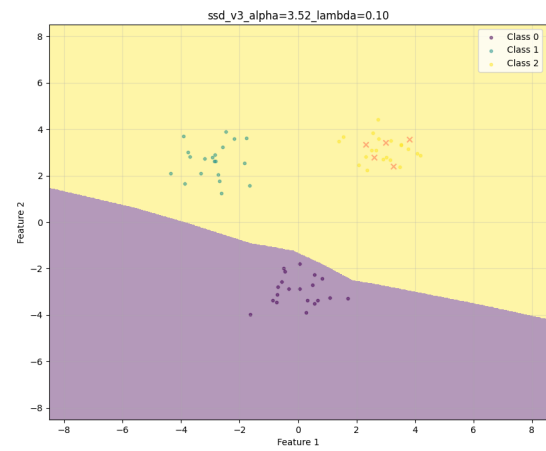
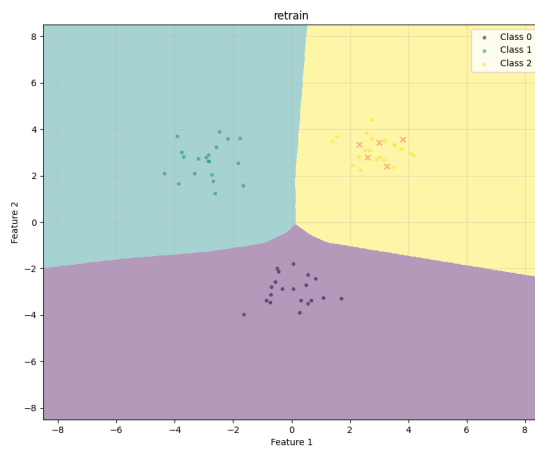
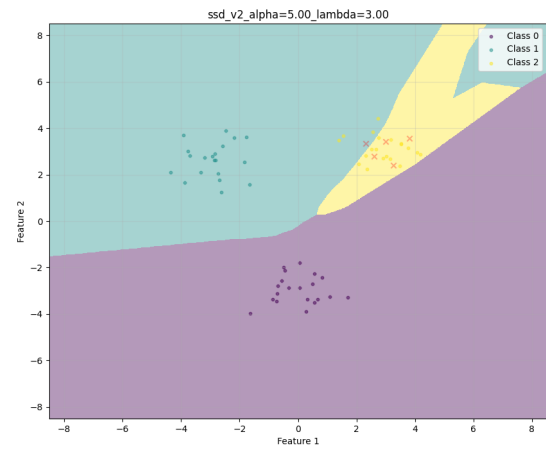
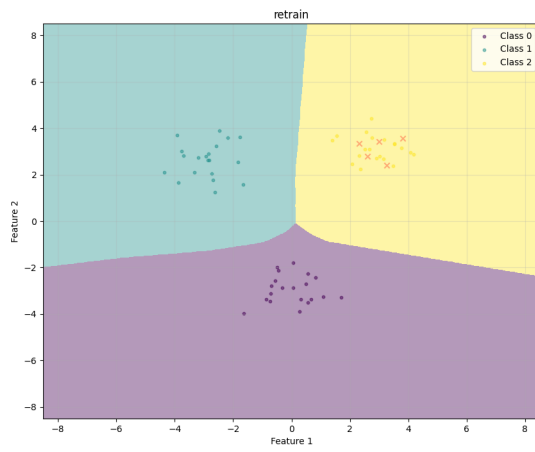


Findings

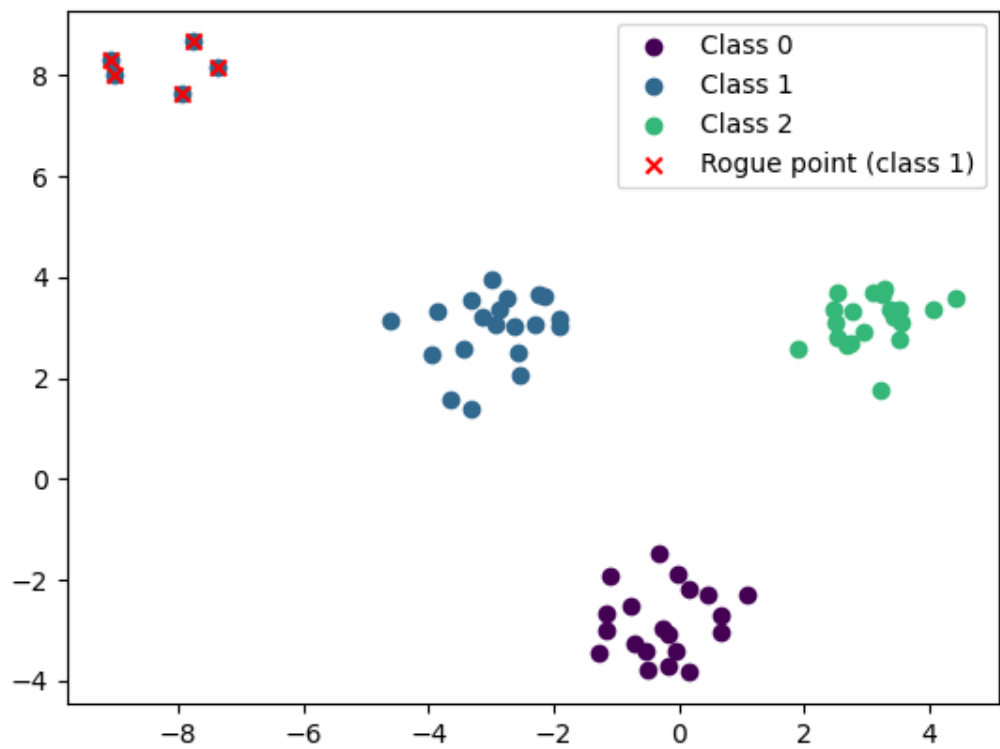
Unlearned

SSD

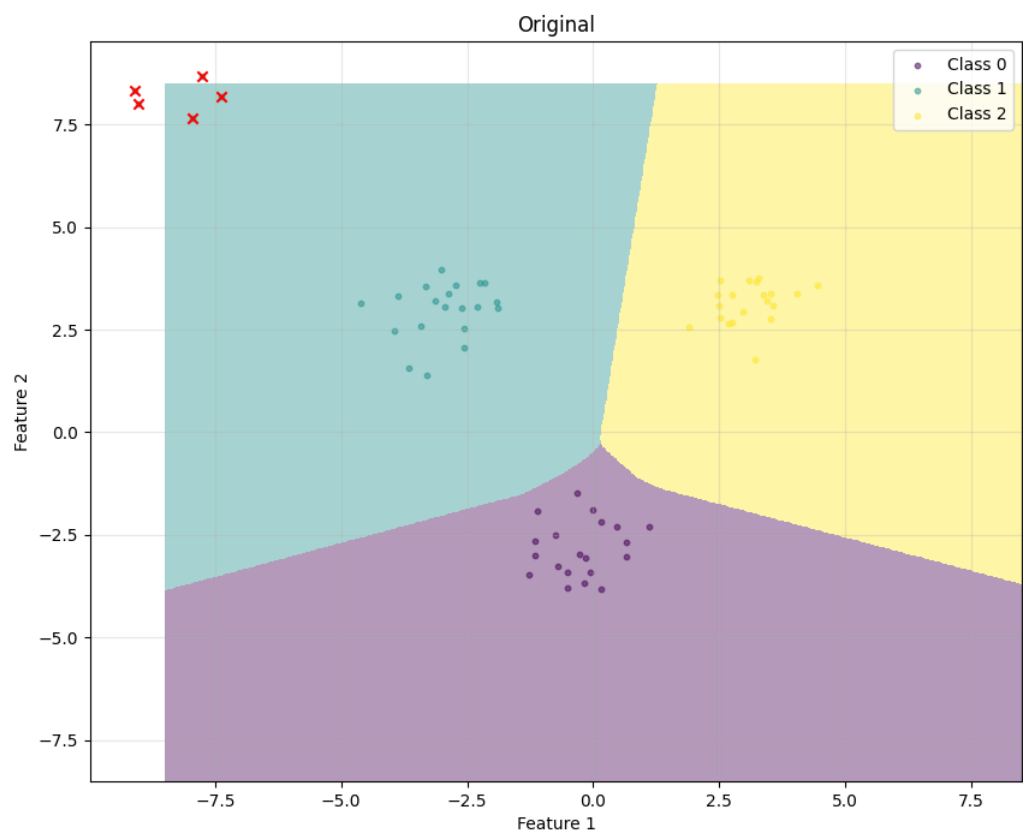




4. Rogue point far away from its centroid, but probably in the same decision boundary



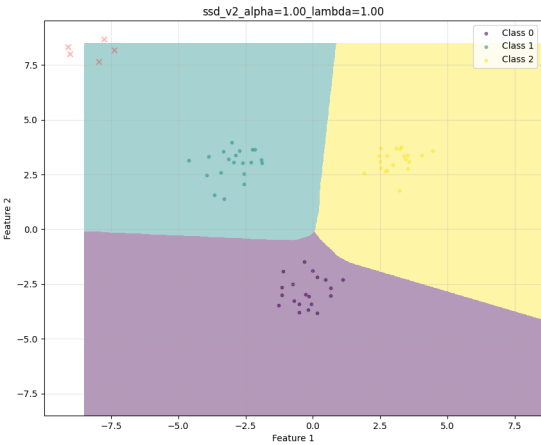
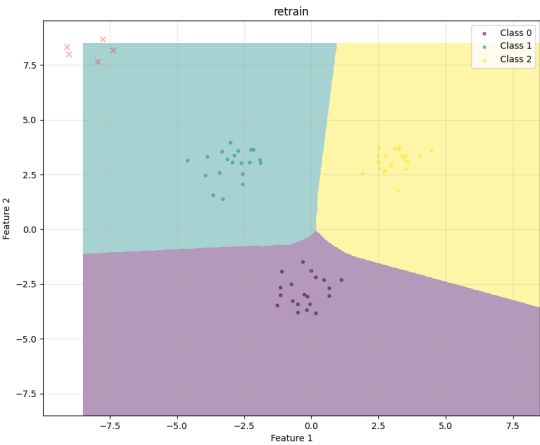
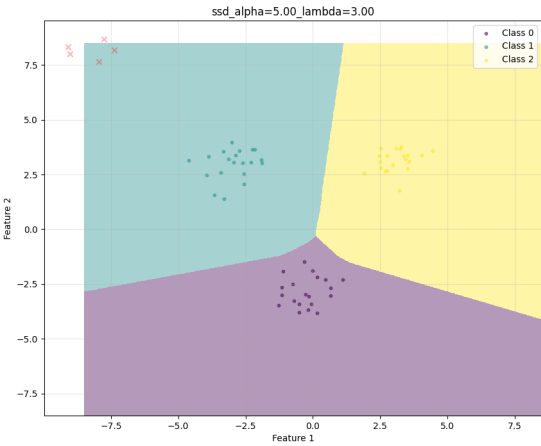
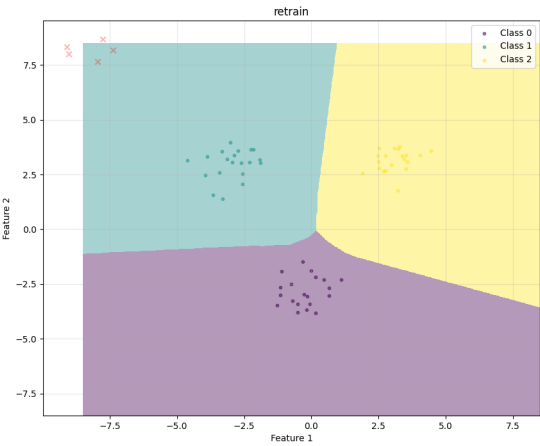
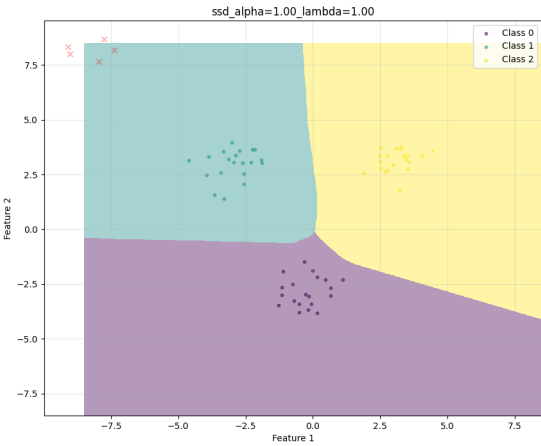
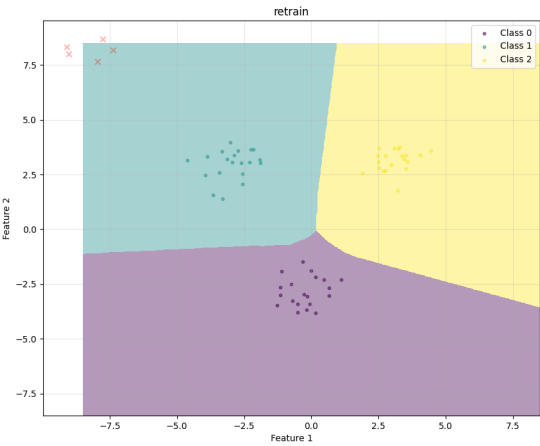
Original model for the dataset

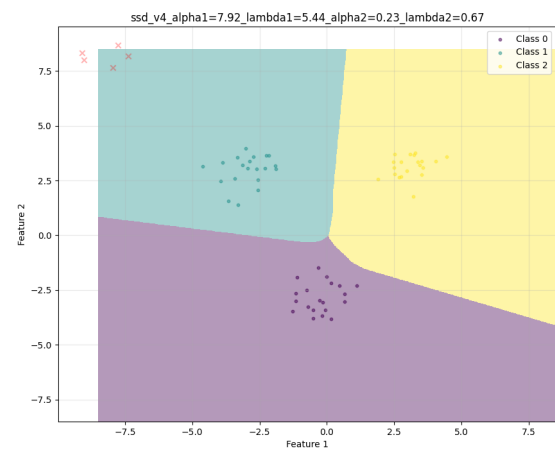
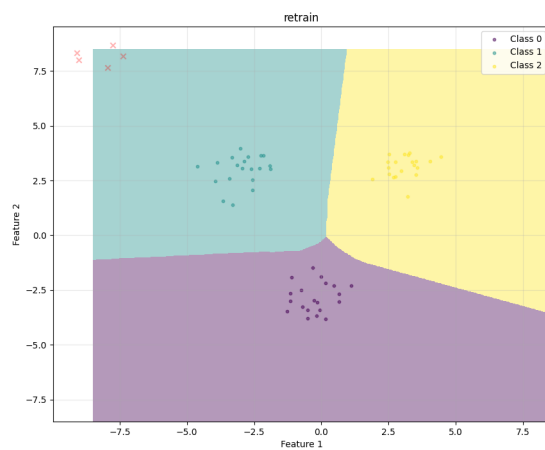
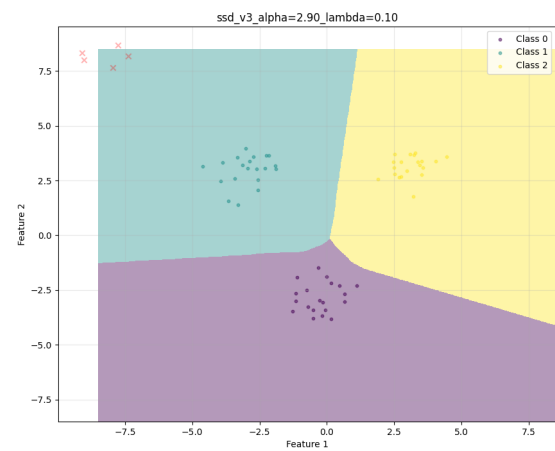
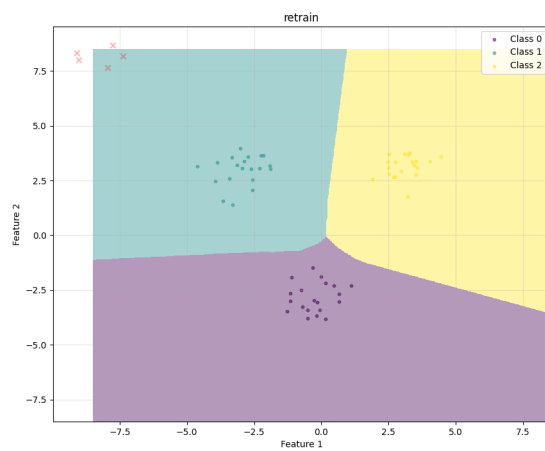
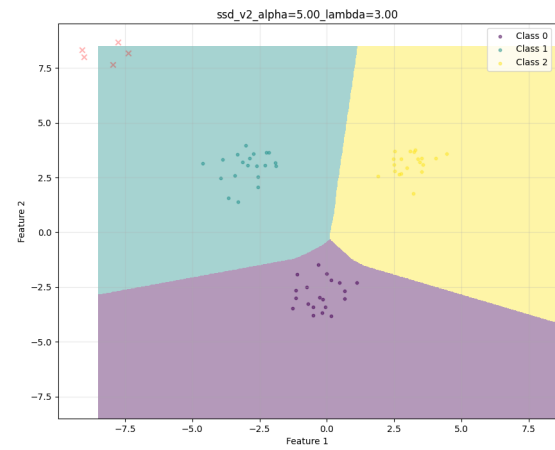
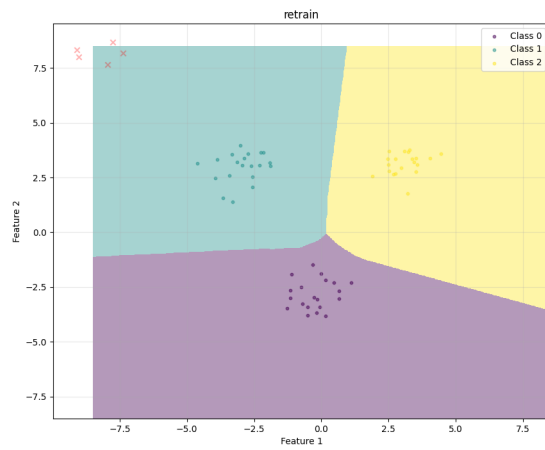


Findings

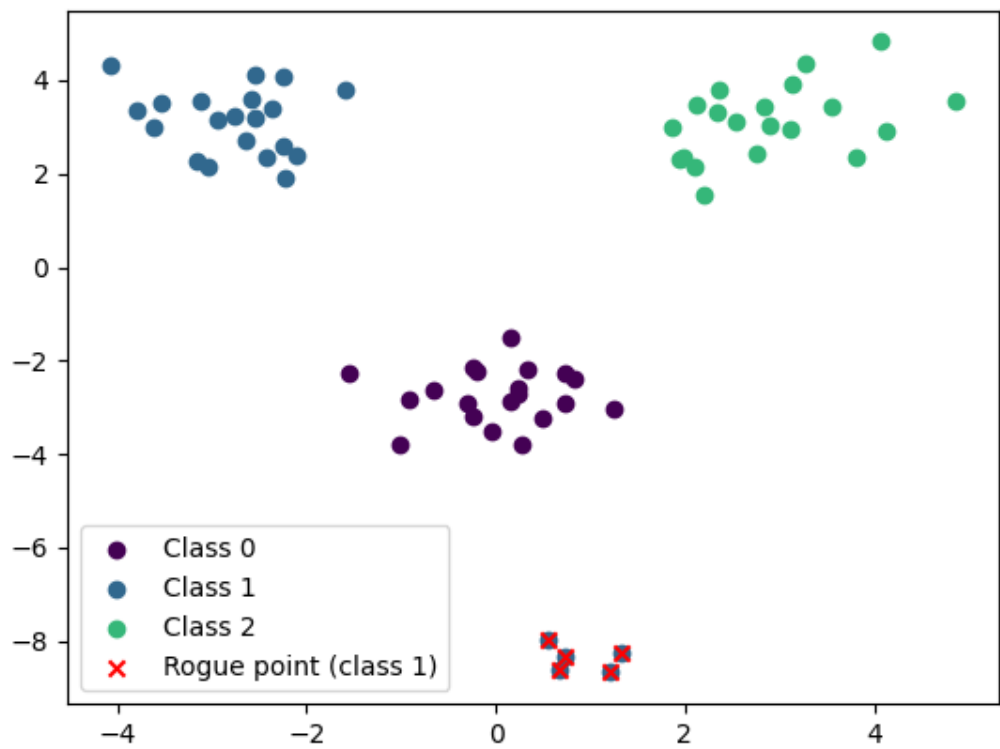
Unlearned

SSD

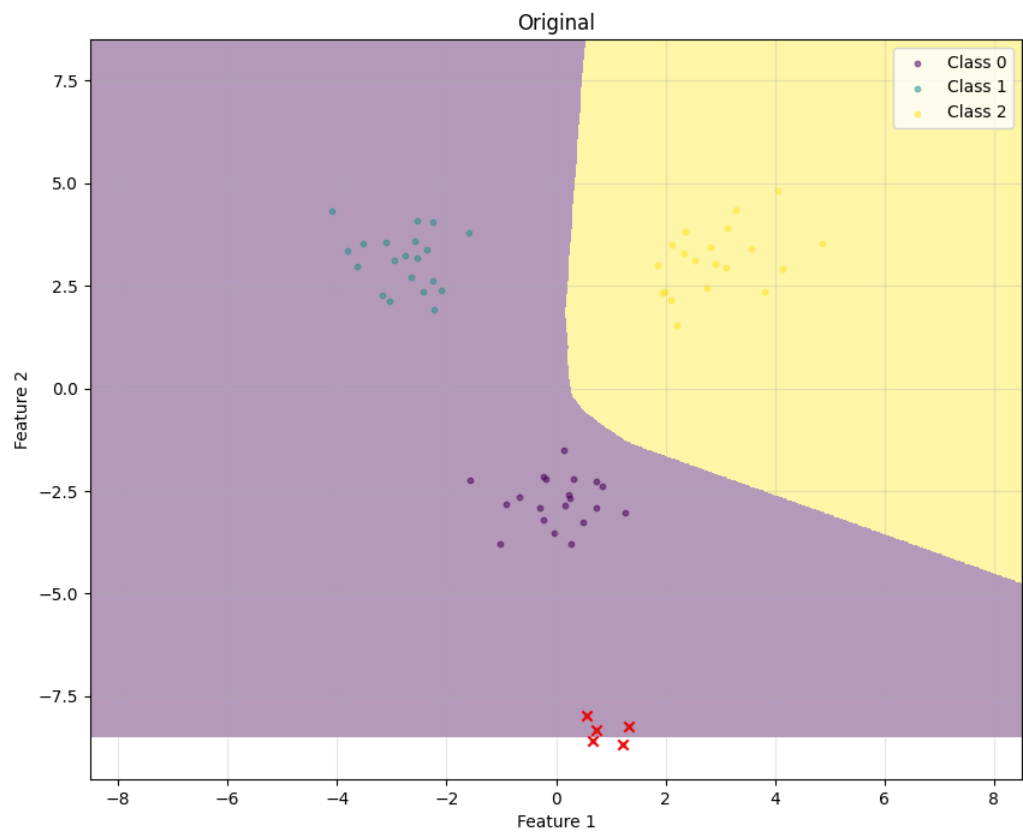




5. Rogue point far away from its centroid, but probably in the same decision boundary

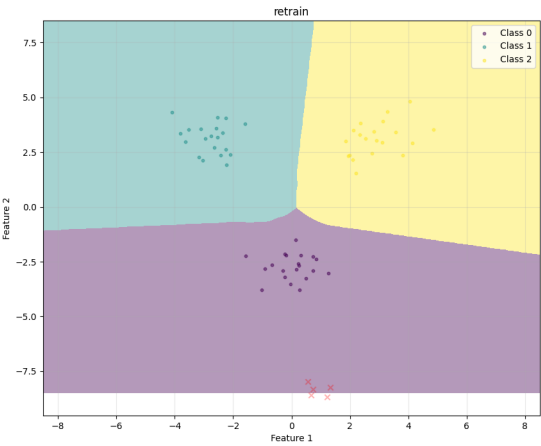


Original model for the dataset



Findings

Unlearned



SSD

